



Course Code	: 2104CS352	Date	: 13-05-2025
Course Name	: Internet of Things	Duration	: 150 Minutes
		Total Marks	: 70

Instructions:

1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

- Q.1 (A)** List the various application areas of IoT. **4**
- (B)** Explain what “Things” refers to in IoT. **3**

OR

Describe how an embedded system functions as an enabling technology in IoT.

- (C)** Draw and explain the IoT 7-layer stack, focusing on its components. **7**

OR

Explain the levels of an IoT system in detail.

- Q.2 (A)** Explain the difference between active and passive components in an electronics system. **4**
- (B)** State Ohm’s Law as it applies to an electrical circuit. **3**

OR

List the SI units for resistor, voltage, and current.

- (C)** Calculate the equivalent resistance of the given circuit. **7**

OR

Determine the equivalent resistance of the provided circuit.

- Q.3 (A)** Describe the working principle of a Light Dependent Resistor (LDR). **4**
- (B)** Explain the difference between sensors and actuators. **3**

OR

Describe how a relay works, including its pin diagram.

- (C)** Write an Arduino code to measure distance using an ultrasonic sound sensor. **7**

OR

Write an Arduino code to detect motion with a PIR motion sensor.

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|------------|------------|--|----------|
| Q.4 | (A) | Explain the basic code structure of Arduino UNO. | 4 |
| | (B) | Write an Arduino code to blink an LED (pin 13) with a 10-second delay. | 3 |

OR

Write an Arduino code to read the status of a slide switch.

- (C)** Draw and explain the block diagram of Arduino UNO in detail. **7**

OR

Explain the purpose of `Serial.begin()`, `Serial.print()`, and `Serial.println()` functions.

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|------------|------------|--|----------|
| Q.5 | (A) | Describe how cloud computing supports IoT. | 4 |
| | (B) | Explain what fog computing is in the context of IoT. | 3 |

OR

Describe the SaaS cloud service model and its use in IoT.

- (C)** Draw and describe a common architecture for an IoT application. **7**

OR

Draw and describe the architecture of a Smart Home application using IoT.
